

Multivariate Analysis: Final Test Bank

Elementary Concentration and Correlation Inequalities

Conventions. $Z \sim N(0, 1)$ has CDF Φ ; $\mathbf{g} \sim N(\mathbf{0}, \mathbf{I}_n)$. For a square matrix, $|\mathbf{A}|$ is the determinant and $\mathbf{A} \succ 0$ means symmetric positive definite. var, cov are variance and covariance. The error function is $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} du$. A *Rademacher* variable takes the values $+1$ and -1 with probability $\frac{1}{2}$ each (mean 0, variance 1); the symbol ε_i always denotes independent such variables.

Facts you may cite without proof.

- (F1) $1 + x \leq e^x$ for all $x \in \mathbb{R}$; Cauchy–Schwarz $\mathbb{E}[UV]^2 \leq \mathbb{E}U^2 \mathbb{E}V^2$; Jensen’s inequality.
- (F2) For $X \sim N(0, \sigma^2)$, $\mathbb{E}e^{\lambda X} = e^{\sigma^2 \lambda^2 / 2}$; $\int_{\mathbb{R}} e^{\mu u^2 - u^2/2} \frac{du}{\sqrt{2\pi}} = (1 - 2\mu)^{-1/2}$ for $\mu < \frac{1}{2}$; and $\mathbb{P}(|Z| \leq a) = 2\Phi(a) - 1 = \text{erf}(a/\sqrt{2})$.
- (F3) A density equals the inverse Fourier transform of its characteristic function, and for Gaussian-type integrands one may differentiate under the integral sign.
- (F4) Symmetric matrices are orthogonally diagonalizable.
- (F5) $\varphi(y) = (e^y - 1 - y)/y^2$ is nondecreasing on $\mathbb{R}$ (with $\varphi(0) = \frac{1}{2}$).
- (F6) Union bound: $\mathbb{P}(\bigcup_i A_i) \leq \sum_i \mathbb{P}(A_i)$.

Problem 1 (Markov, Chebyshev, and Cantelli). Let X have mean μ and variance $\sigma^2 < \infty$.

- (a) (Markov) For $Y \geq 0$ and $t > 0$, prove $\mathbb{P}(Y \geq t) \leq \mathbb{E}Y/t$.
- (b) (Chebyshev) Prove $\mathbb{P}(|X - \mu| \geq t) \leq \sigma^2/t^2$.
- (c) (Cantelli) Prove $\mathbb{P}(X - \mu \geq t) \leq \sigma^2/(\sigma^2 + t^2)$ for $t > 0$.

Solution. (a) $t \mathbf{1}_{\{Y \geq t\}} \leq Y$ pointwise; take expectations: $t \mathbb{P}(Y \geq t) \leq \mathbb{E}Y$.

(b) Apply (a) to $(X - \mu)^2$ at level t^2 : $\mathbb{P}(|X - \mu| \geq t) = \mathbb{P}((X - \mu)^2 \geq t^2) \leq \sigma^2/t^2$.

(c) For $u \geq 0$, $\{X - \mu \geq t\} \subseteq \{(X - \mu + u)^2 \geq (t + u)^2\}$, so by (a) $\mathbb{P}(X - \mu \geq t) \leq \frac{\sigma^2 + u^2}{(t + u)^2}$; minimizing over u (at $u = \sigma^2/t$) gives $\sigma^2/(\sigma^2 + t^2)$. $\square$

Problem 2 (The Chernoff method). (a) Show $\mathbb{P}(X \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} \mathbb{E}e^{\lambda X}$.

- (b) For $X \sim N(0, \sigma^2)$, deduce $\mathbb{P}(X \geq t) \leq e^{-t^2/(2\sigma^2)}$.
- (c) For $S_n = \sum_{i=1}^n \varepsilon_i$, prove $\cosh \lambda \leq e^{\lambda^2/2}$ and conclude $\mathbb{P}(S_n \geq t) \leq e^{-t^2/(2n)}$.

Solution. (a) For $\lambda > 0$, $\{X \geq t\} = \{e^{\lambda X} \geq e^{\lambda t}\}$; Markov (Problem 1a) gives $\mathbb{P}(X \geq t) \leq e^{-\lambda t} \mathbb{E}e^{\lambda X}$; take the infimum.

(b) With $\mathbb{E}e^{\lambda X} = e^{\sigma^2 \lambda^2 / 2}$ ((F2)), minimize $e^{-\lambda t + \sigma^2 \lambda^2 / 2}$ at $\lambda^* = t/\sigma^2$ to get $e^{-t^2/(2\sigma^2)}$.

(c) $\cosh \lambda = \sum_k \frac{\lambda^{2k}}{(2k)!} \leq \sum_k \frac{(\lambda^2/2)^k}{k!} = e^{\lambda^2/2}$ (as $(2k)! \geq 2^k k!$). Then $\mathbb{E}e^{\lambda S_n} = (\cosh \lambda)^n \leq e^{n\lambda^2/2}$; minimize at $\lambda^* = t/n$ for $e^{-t^2/(2n)}$. $\square$

Problem 3 (Hoeffding’s lemma). Let $\mathbb{E}X = 0$ and $a \leq X \leq b$ a.s. Prove $\mathbb{E}e^{\lambda X} \leq e^{\lambda^2(b-a)^2/8}$ for all λ .

Solution. If $a = b$, $X \equiv 0$ and both sides are 1; assume $a < b$. Convexity gives, for $x \in [a, b]$, $e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b}$; taking expectations with $\mathbb{E}X = 0$,

$$\mathbb{E}e^{\lambda X} \leq \frac{b e^{\lambda a} - a e^{\lambda b}}{b-a} = e^{\varphi(h)}, \quad h = \lambda(b-a), \quad \theta = \frac{-a}{b-a} \in [0, 1],$$

with $\varphi(h) = -\theta h + \ln(1 - \theta + \theta e^h)$. Then $\varphi(0) = \varphi'(0) = 0$ and $\varphi''(h) = u(1-u) \leq \frac{1}{4}$ for $u = \frac{\theta e^h}{1 - \theta + \theta e^h}$, so by Taylor $\varphi(h) \leq h^2/8$, i.e. $\mathbb{E}e^{\lambda X} \leq e^{\lambda^2(b-a)^2/8}$. $\square$

Problem 4 (Hoeffding’s inequality). Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = 0$, $a_i \leq X_i \leq b_i$.

(a) Prove $P(|\sum_i X_i| \geq t) \leq 2 \exp(-2t^2 / \sum_i (b_i - a_i)^2)$.

(b) If the X_i are i.i.d. with mean μ and $a \leq X_i \leq b$, deduce $P(|\bar{X} - \mu| \geq t) \leq 2e^{-2nt^2/(b-a)^2}$.

Solution. (a) By the Chernoff method (Problem 2a), independence, and Hoeffding's lemma (Problem 3), $P(\sum_i X_i \geq t) \leq \exp(-\lambda t + \frac{\lambda^2}{8} \sum_i (b_i - a_i)^2)$; minimizing at $\lambda^* = 4t / \sum_i (b_i - a_i)^2$ gives $\exp(-2t^2 / \sum_i (b_i - a_i)^2)$. The two-sided bound follows by applying this to $-X_i$ and adding.

(b) Apply (a) to $Y_i = X_i - \mu$ (range $b - a$): $\sum_i Y_i = n(\bar{X} - \mu)$, so $P(|n(\bar{X} - \mu)| \geq nt) \leq 2e^{-2(nt)^2/(n(b-a)^2)} = 2e^{-2nt^2/(b-a)^2}$. $\square$

Problem 5 (Multiplicative Chernoff bound). Let $X_1, \dots, X_n$ be independent, $X_i \in \{0, 1\}$, $P(X_i = 1) = p_i$, $S = \sum_i X_i$, $\mu = ES$. Prove for $\delta > 0$,

$$P(S \geq (1 + \delta)\mu) \leq \left(\frac{e^\delta}{(1 + \delta)^{1 + \delta}} \right)^\mu \leq e^{-\mu\delta^2/3} \quad (0 < \delta \leq 1).$$

Solution. For $\lambda > 0$, $Ee^{\lambda X_i} = 1 + p_i(e^\lambda - 1) \leq e^{p_i(e^\lambda - 1)}$ ((F1)), so $Ee^{\lambda S} \leq e^{\mu(e^\lambda - 1)}$ and $P(S \geq (1 + \delta)\mu) \leq e^{-\lambda(1 + \delta)\mu + \mu(e^\lambda - 1)}$. At $\lambda = \ln(1 + \delta)$ the exponent is $\mu[\delta - (1 + \delta)\ln(1 + \delta)]$, the first bound. For $0 < \delta \leq 1$, $(1 + \delta)\ln(1 + \delta) \geq \delta + \delta^2/3$ gives $\leq e^{-\mu\delta^2/3}$. $\square$

Problem 6 (Bennett's and Bernstein's inequalities). Let $X_1, \dots, X_n$ be independent, $EX_i = 0$, $X_i \leq b$ a.s., $\sigma^2 = \sum_i \text{var}(X_i)$.

(a) (Bennett) Prove $P(\sum_i X_i \geq t) \leq \exp(-\frac{\sigma^2}{b^2} h(\frac{bt}{\sigma^2}))$, $h(u) = (1 + u)\ln(1 + u) - u$.

(b) (Bernstein) Using $h(u) \geq \frac{u^2}{2(1 + u/3)}$, deduce $P(\sum_i X_i \geq t) \leq \exp(-\frac{t^2}{2(\sigma^2 + bt/3)})$.

Solution. (a) For $x \leq b$, $e^{\lambda x} - 1 - \lambda x = (\lambda x)^2 \varphi(\lambda x) \leq \frac{x^2}{b^2} (e^{\lambda b} - 1 - \lambda b)$ since φ is nondecreasing ((F5)). Taking expectations ($EX_i = 0$) and using $1 + y \leq e^y$, $Ee^{\lambda X_i} \leq \exp(\frac{\text{var}(X_i)}{b^2} (e^{\lambda b} - 1 - \lambda b))$; multiplying and applying Chernoff, $P(\sum_i X_i \geq t) \leq \exp(-\lambda t + \frac{\sigma^2}{b^2} (e^{\lambda b} - 1 - \lambda b))$. Minimizing (at $\lambda = \frac{1}{b} \ln(1 + \frac{bt}{\sigma^2})$) gives the exponent $-\frac{\sigma^2}{b^2} h(\frac{bt}{\sigma^2})$.

(b) Substitute $h(u) \geq \frac{u^2}{2(1 + u/3)}$ with $u = bt/\sigma^2$ to get $-\frac{t^2}{2(\sigma^2 + bt/3)}$. $\square$

Problem 7 (Maximal inequality for Gaussian-type variables). Let $X_1, \dots, X_N$ (not necessarily independent) each satisfy $Ee^{\lambda X_i} \leq e^{\sigma^2 \lambda^2/2}$ for all λ .

(a) Show $P(\max_{i \leq N} X_i \geq t) \leq Ne^{-t^2/(2\sigma^2)}$.

(b) Show $E[\max_{i \leq N} X_i] \leq \sigma\sqrt{2 \ln N}$.

Solution. (a) By Chernoff (Problem 2a) and the hypothesis, $P(X_i \geq t) \leq e^{-t^2/(2\sigma^2)}$; the union bound ((F6)) gives $P(\max_i X_i \geq t) \leq Ne^{-t^2/(2\sigma^2)}$.

(b) For $\lambda > 0$, by Jensen ((F1)), $e^{\lambda E \max_i X_i} \leq E \max_i e^{\lambda X_i} \leq \sum_i Ee^{\lambda X_i} \leq Ne^{\sigma^2 \lambda^2/2}$, so $E \max_i X_i \leq \frac{\ln N}{\lambda} + \frac{\sigma^2 \lambda}{2}$; optimize at $\lambda = \sqrt{2 \ln N}/\sigma$. $\square$

Problem 8 (The Paley–Zygmund (second-moment) inequality). Let $X \geq 0$ with $0 < EX^2 < \infty$. Prove for $\theta \in [0, 1]$, $P(X > \theta EX) \geq (1 - \theta)^2 (EX)^2 / EX^2$.

Solution. Split $EX = E[X \mathbf{1}_{\{X \leq \theta EX\}}] + E[X \mathbf{1}_{\{X > \theta EX\}}]$; the first term is $\leq \theta EX$. By Cauchy–Schwarz ((F1)), $E[X \mathbf{1}_{\{X > \theta EX\}}] \leq (EX^2)^{1/2} P(X > \theta EX)^{1/2}$. Thus $(1 - \theta)EX \leq (EX^2)^{1/2} P(X > \theta EX)^{1/2}$; square and rearrange. $\square$

Problem 9 (Sharpness for Rademacher sums). Let $S_n = \sum_{i=1}^n \varepsilon_i$.

(a) Compute $Ee^{\lambda S_n}$ exactly.

(b) Show $\cosh \lambda \leq e^{\lambda^2/2}$ is tight to second order at $\lambda = 0$, so the exponent $t^2/(2n)$ cannot be improved to ct^2/n with $c < \frac{1}{2}$ by the Chernoff method.

Solution. (a) $\mathbb{E}e^{\lambda S_n} = (\mathbb{E}e^{\lambda \varepsilon_1})^n = (\cosh \lambda)^n$.

(b) $\cosh \lambda = 1 + \frac{\lambda^2}{2} + \frac{\lambda^4}{24} + \dots$ and $e^{\lambda^2/2} = 1 + \frac{\lambda^2}{2} + \frac{\lambda^4}{8} + \dots$ agree through λ^2 , so $\ln \cosh \lambda = \frac{\lambda^2}{2}(1 + o(1))$. A bound $\ln \cosh \lambda \leq c\lambda^2$ with $c < \frac{1}{2}$ fails for small λ , so the Chernoff exponent cannot beat the value at $c = \frac{1}{2}$, namely $e^{-t^2/(2n)}$. $\square$

Problem 10 (χ^2 concentration). Let $Z_1, \dots, Z_n$ be i.i.d. $N(0, 1)$, $\chi_n^2 = \sum_i Z_i^2$.

(a) Show $\mathbb{E}e^{\mu \chi_n^2} = (1 - 2\mu)^{-n/2}$ for $\mu < \frac{1}{2}$.

(b) Prove $\mathbb{P}(\chi_n^2 \geq n(1 + \delta)) \leq e^{-n\delta^2/8}$ for $\delta \in (0, 1]$.

Solution. (a) $\mathbb{E}e^{\mu Z_i^2} = (1 - 2\mu)^{-1/2}$ ((F2)); by independence $\mathbb{E}e^{\mu \chi_n^2} = (1 - 2\mu)^{-n/2}$.

(b) Chernoff: $\mathbb{P}(\chi_n^2 \geq n(1 + \delta)) \leq e^{-\mu n(1 + \delta)}(1 - 2\mu)^{-n/2}$. At $\mu = \frac{\delta}{2(1 + \delta)}$ (so $1 - 2\mu = \frac{1}{1 + \delta}$) the exponent is $-\frac{n}{2}(\delta - \ln(1 + \delta)) \leq -\frac{n\delta^2}{8}$, using $\delta - \ln(1 + \delta) \geq \delta^2/4$ on $[0, 1]$. $\square$

Problem 11 (Lower-tail Chernoff bounds). (a) For S as in Problem 5 and $0 < \delta < 1$, prove $\mathbb{P}(S \leq (1 - \delta)\mu) \leq e^{-\mu\delta^2/2}$.

(b) For $\chi_n^2 = \sum_i Z_i^2$ and $\delta \in (0, 1)$, prove $\mathbb{P}(\chi_n^2 \leq n(1 - \delta)) \leq e^{-n\delta^2/4}$, hence $\mathbb{P}(|\chi_n^2 - n| \geq n\delta) \leq 2e^{-n\delta^2/8}$.

Solution. (a) For $\lambda > 0$, $\mathbb{P}(S \leq (1 - \delta)\mu) \leq e^{\lambda(1 - \delta)\mu} \mathbb{E}e^{-\lambda S} \leq e^{\mu[\lambda(1 - \delta) + e^{-\lambda} - 1]}$. At $\lambda = -\ln(1 - \delta)$ the exponent is $\mu[-\delta - (1 - \delta)\ln(1 - \delta)] \leq -\mu\delta^2/2$ (using $(1 - \delta)\ln(1 - \delta) \geq -\delta + \delta^2/2$).

(b) For $s > 0$, $\mathbb{P}(\chi_n^2 \leq n(1 - \delta)) \leq e^{sn(1 - \delta)}(1 + 2s)^{-n/2}$ (Problem 10a, $\mu = -s$). At $1 + 2s = (1 - \delta)^{-1}$ the exponent is $\frac{n}{2}(\delta + \ln(1 - \delta)) \leq -\frac{n\delta^2}{4}$. With Problem 10(b), $\mathbb{P}(|\chi_n^2 - n| \geq n\delta) \leq e^{-n\delta^2/8} + e^{-n\delta^2/4} \leq 2e^{-n\delta^2/8}$. $\square$

Problem 12 (The Johnson–Lindenstrauss lemma). Let $x_1, \dots, x_N \in \mathbb{R}^D$, $\varepsilon \in (0, 1)$, $\mathbf{A}$ an $m \times D$ matrix with i.i.d. $N(0, 1)$ entries, $f(x) = \mathbf{A}x/\sqrt{m}$. Show that if $m \geq 24\varepsilon^{-2} \ln N$ then with positive probability $(1 - \varepsilon)\|x_i - x_j\|^2 \leq \|f(x_i) - f(x_j)\|^2 \leq (1 + \varepsilon)\|x_i - x_j\|^2$ for all i, j .

Solution. Fix $i \neq j$, $u = x_i - x_j \neq 0$. The rows $r_1, \dots, r_m$ of $\mathbf{A}$ are i.i.d. $N(\mathbf{0}, \mathbf{I}_D)$, so $\langle r_k, u \rangle / \|u\|$ are i.i.d. $N(0, 1)$ and $\|f(x_i) - f(x_j)\|^2 / \|u\|^2 = \frac{1}{m} \sum_k (\langle r_k, u \rangle / \|u\|)^2 = \chi_m^2 / m$. By Problem 11(b), $\mathbb{P}(|\chi_m^2 / m - 1| \geq \varepsilon) \leq 2e^{-m\varepsilon^2/8}$. Union over $< N^2$ pairs ((F6)): the failure probability is $\leq N^2 \cdot 2e^{-m\varepsilon^2/8} \leq 2N^2 e^{-3 \ln N} = 2/N < 1$ for $N \geq 3$, so a suitable $\mathbf{A}$ exists. $\square$

Problem 13 (Gaussian measure of a disk; two easy instances of the GCI). Let μ be a standard Gaussian measure.

(a) For $G \sim N(\mathbf{0}, \mathbf{I}_2)$, show $\mu(\|x\| \leq R) = 1 - e^{-R^2/2}$.

(b) (Nested) For $K = \{\|x\| \leq R_1\} \subseteq L = \{\|x\| \leq R_2\}$, show $\mu(K \cap L) \geq \mu(K)\mu(L)$; evaluate at $R_1^2 = 2 \ln 2$, $R_2^2 = 2 \ln 4$.

(c) (Independent) For $N(\mathbf{0}, \mathbf{I}_3)$, $K = \{|x_1| \leq a\}$, $L = \{x_2^2 + x_3^2 \leq R^2\}$, show $\mu(K \cap L) = \mu(K)\mu(L)$.

Solution. (a) Polar coordinates: $\mu(\|x\| \leq R) = \int_0^{2\pi} \int_0^R \frac{1}{2\pi} e^{-r^2/2} r \, dr \, d\theta = 1 - e^{-R^2/2}$.

(b) $K \subseteq L \Rightarrow K \cap L = K$, so $\mu(K \cap L) = \mu(K) \geq \mu(K)\mu(L)$ as $\mu(L) \leq 1$. Here $\mu(K) = \frac{1}{2}$, $\mu(L) = \frac{3}{4}$, and $\frac{1}{2} \geq \frac{3}{8}$.

(c) The density factors across $\{1\}$ and $\{2, 3\}$ and $K \cap L$ is the corresponding product set, so by Fubini $\mu(K \cap L) = \mu(K)\mu(L)$ ($\mu(K) = \text{erf}(a/\sqrt{2})$, $\mu(L) = 1 - e^{-R^2/2}$). $\square$

Problem 14 (The bivariate Gaussian correlation inequality, via Plackett). Let $X \sim N(\mathbf{0}, \mathbf{C}_\rho)$, $\mathbf{C}_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $|\rho| < 1$, $K = \{|x_1| \leq a\}$, $L = \{|x_2| \leq a\}$; Φ_2, ϕ_2 the bivariate normal CDF/density.

(a) (Plackett) Prove $\partial_\rho \Phi_2(h, k; \rho) = \phi_2(h, k; \rho)$.

(b) With $P(\rho) = \mu(K \cap L)$, prove $P'(\rho) = \frac{1}{\pi\sqrt{1-\rho^2}}(e^{-a^2/(1+\rho)} - e^{-a^2/(1-\rho)})$.

(c) Conclude $\mu(K \cap L) > \mu(K)\mu(L)$ for $\rho \neq 0$, with equality iff $\rho = 0$.

Solution. (a) Writing ϕ_2 as the inverse Fourier transform of $e^{-\frac{1}{2}(s^2+2\rho st+t^2)}$ ((F3)), ∂_ρ brings down $-st$ and $\partial_h\partial_k$ brings down $(-is)(-it) = -st$, so $\partial_\rho\phi_2 = \partial^2\phi_2/\partial h\partial k$. Integrating over $(-\infty, h] \times (-\infty, k]$ (decay at $-\infty$) gives $\partial_\rho\Phi_2 = \phi_2$.

(b) Inclusion-exclusion: $P(\rho) = \Phi_2(a, a) - \Phi_2(-a, a) - \Phi_2(a, -a) + \Phi_2(-a, -a)$; differentiate and use (a). The exponent $Q = \frac{h^2-2\rho hk+k^2}{2(1-\rho^2)}$ gives $Q(\pm a, \pm a) = \frac{a^2}{1+\rho}$, $Q(\pm a, \mp a) = \frac{a^2}{1-\rho}$; with prefactor $\frac{1}{2\pi\sqrt{1-\rho^2}}$ (each value twice), $P'(\rho) = \frac{1}{\pi\sqrt{1-\rho^2}}(e^{-a^2/(1+\rho)} - e^{-a^2/(1-\rho)})$.

(c) For $\rho \in (0, 1)$, $\frac{a^2}{1+\rho} < \frac{a^2}{1-\rho}$ so $P'(\rho) > 0$; P is even (reflect $x_2 \mapsto -x_2$), so it increases on $[0, 1)$ from $P(0) = (2\Phi(a) - 1)^2 = \mu(K)\mu(L)$. Hence $\mu(K \cap L) > \mu(K)\mu(L)$ for $\rho \neq 0$. $\square$

Problem 15 (How slack is the inequality? Small boxes). In the setting of Problem 14, show that for fixed $\rho \neq 0$, $\frac{\mu(K \cap L)}{\mu(K)\mu(L)} \xrightarrow{a \rightarrow 0} (1 - \rho^2)^{-1/2} > 1$.

Solution. By Problem 14, $\mu(K \cap L) - \mu(K)\mu(L) = \int_0^{|\rho|} P'(s)ds$. Expanding $e^{-a^2/(1\pm s)} = 1 - \frac{a^2}{1\pm s} + O(a^4)$ gives $e^{-a^2/(1+s)} - e^{-a^2/(1-s)} = \frac{2a^2s}{1-s^2} + O(a^4)$, so $\int_0^{|\rho|} \frac{2a^2s}{\pi(1-s^2)^{3/2}}ds = \frac{2a^2}{\pi}((1-\rho^2)^{-1/2} - 1) + O(a^4)$. Since $\mu(K)\mu(L) = (2\Phi(a) - 1)^2 = \frac{2a^2}{\pi} + O(a^4)$, dividing gives $\mu(K \cap L)/(\mu(K)\mu(L)) \rightarrow (1-\rho^2)^{-1/2}$. $\square$